Unit: Unit 8: Quadratics
Topic: Systems of Quadratic Equations

Name: ________________
Date: ___________

Quiz (Habanero)

Q1. Solve the system: $x^2 + y^2 = 10$, $x^2 - y^2 = 2$
(A) $x = 2, y = 2$
(B) $x = 2, y = -2$
(C) $x = -2, y = 2$
(D) $x = -2, y = -2$
(E) $x = 2, y = 0$

Q2. Graphically, the solution to $y = x^2 - 4$ and $y = -x^2 + 4$ is
(A) Two intersecting lines
(B) Two parallel lines
(C) One line and one parabola
(D) Two intersecting parabolas
(E) No intersection

Q3. The system $x^2 + 2y^2 = 8$, $2x^2 + y^2 = 9$ has
(A) One solution
(B) Two solutions
(C) Three solutions
(D) Four solutions
(E) Infinite solutions

Q4. Temperature in a city is modeled by $T(t) = t^2 - 4t + 10$ and humidity by $H(t) = -t^2 + 4t + 5$. When do they intersect?
(A) $t = 1$
(B) $t = 2$
(C) $t = 3$
(D) $t = 4$
(E) $t = 5$

Q5. The quadratic equations $y = x^2 - 2x + 1$ and $y = x^2 + 2x - 3$
(A) Have one common solution
(B) Have two common solutions
(C) Have no common solutions
(D) Are parallel
(E) Are the same equation

Q6. Rainfall in a region is given by $R(x) = x^2 - 3x + 2$. When $R(x) = 5$,
(A) $x = 1$ or $x = 4$
(B) $x = 2$ or $x = 3$
(C) $x = -1$ or $x = 5$
(D) $x = 2$ or $x = -1$
(E) $x = -2$ or $x = 3$

Q7. Solve for y when $x^2 + y^2 = 16$ and $x^2 - y^2 = -4$
(A) $y = pm2sqrt3$
(B) $y = pm2sqrt2$
(C) $y = pm4$
(D) $y = pm2$
(E) $y = 0$

Q8. An ecosystem's population growth is modeled by $P(t) = t^2 + 2t + 1$ and its food supply by $F(t) = -t^2 + 2t - 3$. When does the population outgrow its food supply?
(A) $t = 1$
(B) $t = 2$
(C) $t = 3$
(D) $t = 4$
(E) $t = 5$

Open Ended Questions (FRQ)

Q9. Solve the system $x^2 + y^2 = 20$, $x^2 - y^2 = 4$. Show all your work.

Q10. A water balloon toss game's trajectory is given by $y = x^2 - 4x + 5$. When $y = 0$, find x. Explain your method.

Chef's Tips

- Make sure students show their work for FRQs
- Emphasize the importance of checking solutions in the original equations
- Use graphing tools to visualize the systems of quadratic equations if necessary